

Dear Hiring Manager,

I am writing to apply for the position of **Specialist – ADAS Testing using IPG CarMaker**. I hold a Ph.D. in Mechanical Engineering from NIT Warangal and have professional experience as a Robotics Software Engineer, along with my current role as an Assistant Professor in Robotics and Automation. My background combines rigorous research and hands-on industry experience in **autonomous mobile robotics, ROS 2, sensor fusion, localization, planning, control, simulation, and experimental validation**.

I have extensive practical experience in the **development, integration, simulation, testing, and experimental validation of autonomous mobile robot systems**. During my Ph.D., I worked on coverage path planning for mobile robots operating in complex environments, including flat and uneven terrains. I developed improved variants of the **NSGA-II algorithm for multi-objective autonomous navigation**, considering objectives such as travel distance, number of turns, energy efficiency, terrain-related cost, and obstacle avoidance. My research on multi-objective coverage path planning was published in *Robotica* by Cambridge University Press: <https://doi.org/10.1017/S0263574724000377>.

My research was not limited to simulation. I experimentally validated autonomous navigation and coverage path planning on a **differential-drive GoPiGo3 mobile robot using Raspberry Pi**, integrating planning, localization, sensor fusion, and control with a physical robotic platform. This experience has given me a strong understanding of the importance of systematic testing and validation of autonomous systems under real-world conditions.

A major area of my expertise is **sensor fusion and localization for autonomous mobile robots using ROS 2**. I have worked with multiple sensors, including IMU, wheel odometry, infrared sensors, drag wheel, and barometer, and developed adaptive Kalman filtering and multi-sensor fusion approaches for reliable pose estimation. I have also implemented motion and PD-based control approaches and validated the resulting system on physical robotic platforms.

I also have strong experience in **simulation-based robotic system development and testing using ROS 2, Webots, and Gazebo**, along with C, C++, Python, MATLAB, and Git/GitHub. My experience involves developing algorithms, integrating sensors and system components, evaluating system behavior in simulation, analyzing results, and subsequently validating the developed approach on real robotic hardware.

Although I have **not yet worked directly with IPG CarMaker**, I believe my extensive experience with ROS 2-based simulation, autonomous navigation, sensor fusion, state estimation, control, and experimental validation provides a strong transferable foundation for ADAS testing. I am highly motivated to learn IPG CarMaker and apply my existing experience to automotive simulation, scenario-based testing, verification, and validation of ADAS functions.

Prior to joining academia, I worked as a **Robotics Software Engineer at WCB Robotics India Pvt. Ltd.**, where I developed algorithms for localization, SLAM, sensor fusion, control systems, robot kinematics, and coverage path planning for real-world façade-cleaning robotic systems. I worked on pose localization using adaptive Kalman filtering and multi-sensor fusion involving IR, IMU, odometry, drag wheel, and barometer data. This experience provided me with practical exposure to the complete development cycle, from algorithm development and simulation to system integration and validation on physical robotic platforms.

My research experience with **multi-objective optimization and NSGA-II** has also strengthened my ability to analyze autonomous-system performance under competing requirements. In autonomous systems, decisions often need to balance safety, efficiency, trajectory quality, obstacle avoidance, motion smoothness, and operational constraints. I believe this perspective is valuable for ADAS testing, where different driving scenarios and system conditions must be evaluated systematically to assess performance and robustness.

I am particularly interested in this position because it provides an opportunity to transition my strong background in **autonomous robotics testing and validation into automotive ADAS development**. I am confident that my experience in ROS 2, simulation, sensor fusion, localization, planning, control, algorithm development, and physical-system validation, combined with my willingness to quickly learn IPG CarMaker, would allow me to contribute effectively to your team.

I would welcome the opportunity to discuss how my research background, industrial robotics experience, and hands-on expertise in autonomous system development and validation can contribute to your ADAS testing team.

Thank you for considering my application.

Sincerely,

Dr. Monex Sharma

Assistant Professor | Robotics Software Engineer

Ph.D., NIT Warangal

C/C++ | Python | ROS 2 | Sensor Fusion | Localization | Autonomous Navigation | Control Systems | Simulation & Validation